Freddy Shaikh

SENIOR FULL STACK SOFTWARE ENGINEER · AI & AGENTIC SYSTEMS

Toronto, Canada

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Summary

Senior Full Stack Software Engineer with 7+ years designing, building, and operating cloud-native distributed systems on AWS, and the designated technical lead for AI-driven engineering acceleration at Morningstar. Owns features end-to-end across .NET and Python backends, RESTful and GraphQL APIs, and Vue front ends, deployed through containerized CI/CD and Infrastructure as Code. Built the team's agentic engineering harness - reusable skill files, scoped context, token budgets, and operating-mode guardrails - and mentored senior and principal engineers onto it. AWS Certified Solutions Architect - Professional and Machine Learning Engineer. Self-taught in programming, Japanese, and French.

Skills

Languages	Python, Go, Java, C#, TypeScript, JavaScript, SQL, PowerShell, Bash, HTML, CSS, Gremlin (Graph Query), VBA
Cloud & Infrastructure	AWS (Lambda, ECS, S3, CloudFront, Bedrock, Polly, SNS, IAM), GCP, Docker, Kubernetes, Terraform, Infrastructure as Code, CI/CD (Harness, Jenkins, GitHub Actions)
Backend & APIs	Microservices, Distributed Systems, Event-Driven Architecture, Serverless, REST, GraphQL, Kafka, .NET, Spring Boot, Knowledge Graphs
Frontend	Vue 3, TypeScript, SCSS/Sass, Browser Extensions, Responsive Web, SEO
AI & Agentic	LLM Orchestration, Amazon Bedrock, MCP (Model Context Protocol) Servers, Agentic Coding Harnesses, Autonomous Agent Workflows, Prompt & Context Engineering, Token Budgeting, Agent Guardrails, SSML / Text-to-Speech
Practice & Delivery	Agile, Scrum, Technical Design & Architecture Review, Code Review, TDD, Unit / Integration / API Testing, Playwright, Postman, Jupyter, Tableau, Observability (Splunk, SNS), Jira, Confluence, Mentorship, Technical SME

Experience

Morningstar

Toronto, Canada

SENIOR FULL STACK SOFTWARE ENGINEER

January 2025 - PRESENT

- Designated technical lead for AI-driven engineering acceleration: architected the team's AI-assisted engineering harness (reusable skill files, scoped context and token budgets, operating-mode guardrails) and mentored senior and principal engineers onto it as their default development workflow.
- Technical SME for MCP (Model Context Protocol) server architecture, driving it from proposal and technical design review through production release.
- Owned end-to-end delivery of an AI audio product feature: .NET microservices on AWS Bedrock and Amazon Polly for LLM-driven SSML speech synthesis, plus the Vue 3 front end shipping transcript and smart-summary formats.
- Built a GraphQL API over a Gremlin knowledge graph to model and query concept relationships, improving content discoverability across the platform.
- Designed and deployed Harness CI/CD pipelines in YAML for AWS Lambda and ECS deploys, reducing deployment time.
- Built an automated Playwright test suite covering smoke, unit, and integration tests.
- Architected RESTful APIs powering a Vue-based browser extension, enabling real-time communication between the front end and distributed .NET Lambda back-end services.
- Led the Asian/Pacific Islander Association Employee Resource Group, championing inclusion across the engineering organization.

Santoku Corporation / Rimm.ai

Toronto, Canada

SOFTWARE ENGINEER

August 2019 - January 2025

- Led the rearchitecture of a monolithic on-premises application into AWS cloud microservices, containerized with Docker and orchestrated on Kubernetes for independent deployability, scalability, and fault tolerance.
- Owned custom haptic VR training systems for enterprise clients end-to-end as primary technical decision-maker, from requirements gathering and prototyping through delivery, client training, and support.
- Refactored a legacy codebase to an object-oriented architecture to increase reuse and reduce defect surface area.

Japan Exchange Teaching (JET) Program

Tokyo, Japan

ENGLISH LANGUAGE TEACHER

August 2018 - August 2019

- Delivered English instruction to over 1,000 elementary and 500 middle school students, adapting curriculum using self-taught Japanese (JLPT N2).

Projects

Agentic Engineering Harness (GitHub)

- Designed a provider-neutral harness for autonomous coding agents, portable across GitHub Copilot, Claude Code, and Codex: an idempotent PowerShell installer that projects one skill format into each host's layout, operating modes for supervised vs. unattended posture, a recoverability-over-approval model for overnight runs, and a usage-limit-aware queue that resumes work after rate-limit resets.
- Concurrency and recovery model for parallel agents: exclusive working-tree claims with heartbeats and stale-claim takeover, git-worktree isolation, and compare-and-swap writes to shared state - enforced by a pre-commit hook that rejects an unclaimed agent write, with task state journaled before each risky step so an interrupted run resumes deterministically.

Portfolio Site & Resume Pipeline (freddyshaikh.com)

- Designed a single-source-of-truth publishing pipeline: Python generators cascade one LaTeX source into the website (JavaScript, SCSS, AWS CloudFront), the PDF, and ATS / JSON Resume / LinkedIn exports, enforced by a GitHub Actions workflow with drift detection and Playwright browser tests.

Certifications & Awards

2026	**AWS Certified Machine Learning Engineer - Associate**, AWS Training and Certification	*Toronto, Canada*
2025	**AWS Certified AI Practitioner**, AWS Training and Certification	*Toronto, Canada*
2025	**AWS Certified Developer - Associate**, AWS Training and Certification	*Toronto, Canada*
2024	**Google Cloud Certified: Professional Cloud Architect**, Google Cloud Certification	*Toronto, Canada*
2023	**AWS Certified Solutions Architect - Professional**, AWS Training and Certification	*Toronto, Canada*
2019	**Japanese-Language Proficiency Test (JLPT) - N2**, Japanese Language Proficiency Test	*Tokyo, Japan*

Education

McMaster University *Hamilton, Canada*

BACHELOR OF HONORS SCIENCE - KINESIOLOGY *September 2014 - April 2018*